JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Groups (2005–2014)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
16	75 M	48 Min	Classic Paper Pattern (2

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	16	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q34 • ID: JAM_2014_Q34) MCQ +2 M Neg: −0.67

Q.34 Which one of the following conditions on a group G implies that G is abelian?

(A) The order of G is p^3 for some prime p

(B) Every proper subgroup of G is cyclic

(C) Every subgroup of G is normal in G

(D) The function $f: G \to G$, defined by $f(x) = x^{-1}$ for all $x \in G$, is a homomorphism

Question 2 (JAM 2014 • Q31 • ID: JAM_2014_Q31) MCQ +2 M Neg: −0.67

Q.31 Let S_n be the group of all permutations on the set $\{1, 2, \ldots, n\}$ under the composition of mappings. For $n > 2$, if H is the smallest subgroup of S_n containing the transposition $(1, 2)$ and the cycle $(1, 2, \ldots, n)$, then

(A) $H = S_n$

(B) H is abelian

(C) the index of H in S_n is 2

(D) H is cyclic

Question 3 (JAM 2014 • Q30 • ID: JAM_2014_Q30) MCQ +2 M Neg: −0.67

Q.30 Let G be a cyclic group of order 24. The total number of group isomorphisms of G onto itself is

(A) 7 (B) 8 (C) 17 (D) 24

Question 4 (JAM 2014 • Q7 • ID: JAM_2014_Q7) MCQ +1 M Neg: −0.33

Q.7 Let G be a group of order 17. The total number of non-isomorphic subgroups of G is

(A) 1 (B) 2 (C) 3 (D) 17

Question 5 (JAM 2013 • Q8 • ID: JAM_2013_Q8) MCQ +2 M Neg: −0.67

Let p be a prime number. Let G be the group of all 2×2 matrices over $\mathbb{Z}_p$ with determinant 1 under matrix multiplication. Then the order of G is

(A) $(p-1)p(p+1)$
(B) $p^2(p-1)$
(C) p^3
(D) $p^2(p-1)+p$

Question 6 (JAM 2012 • Q8 • ID: JAM_2012_Q8) MCQ +6 M Neg: −2.0

Q.8 Consider the quotient group $\mathbb{Q}/\mathbb{Z}$ of the additive group of rational numbers. The order of the element $\frac{2}{3}+\mathbb{Z}$ in $\mathbb{Q}/\mathbb{Z}$ is

(A) 2 (B) 3 (C) 5 (D) 6

Question 7 (JAM 2011 • Q10 • ID: JAM_2011_Q10) MCQ +6 M Neg: −2.0

Q.10 For $n \in \mathbb{N}$, let $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$. Then the number of units of $\mathbb{Z}/11\mathbb{Z}$ and $\mathbb{Z}/12\mathbb{Z}$, respectively, are

(A) 11, 12 (B) 10, 11 (C) 10, 4 (D) 10, 8

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Question 8 (JAM 2011 • Q9 • ID: JAM_2011_Q9) MCQ +6 M Neg: −2.0

Q.9 Let G denote the group of all 2×2 invertible matrices with entries from $\mathbb{R}$. Let

$$H_1=\{A\in G:\ \det(A)=1\}\ \text{and}\ \ H_2=\{A\in G: A \text{ is upper triangular}\}.$$

Consider the following statements:

P : H_1 is a normal subgroup of G

Q : H_2 is a normal subgroup of G.

Then

(A) Both P and Q are true
(B) P is true and Q is false
(C) P is false and Q is true
(D) Both P and Q are false

Question 9 (JAM 2010 • Q12 • ID: JAM_2010_Q12) MCQ +6 M Neg: −2.0

Q.12 The number of 2×2 matrices over $\mathbf{Z}_3$ (the field with three elements) with determinant 1 is

(A) 24 (B) 60 (C) 20 (D) 30

Question 10 (JAM 2010 • Q3 • ID: JAM_2010_Q3) MCQ +6 M Neg: −2.0

Q.3 The number of elements of S_5 (the symmetric group on 5 letters) which are their own inverses equals

(A) 10 (B) 11 (C) 25 (D) 26

Question 11 (JAM 2009 • Q5 • ID: JAM_2009_Q5) MCQ +6 M Neg: −2.0

Q.5 The number of all subgroups of the group $(Z_{60},+)$ of integers modulo 60 is

(A) 2 (B) 10 (C) 12 (D) 60

Question 12 (JAM 2008 • Q15 • ID: JAM_2008_Q15) MCQ +6 M Neg: −2.0

Q.15 Let G be a finite group and H be a normal subgroup of G of order 2. Then the order of the center of G is

(A) 0 (B) 1 (C) an even integer ≥ 2 (D) an odd integer ≥ 3

Question 13 (JAM 2008 • Q1 • ID: JAM_2008_Q1) MCQ +6 M Neg: −2.0

Q.1 The least positive integer n, such that $\begin{pmatrix} \cos\frac{\pi}{4} & \sin\frac{\pi}{4} \\ -\sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{pmatrix}^n$ is the identity matrix of order 2, is

(A) 4 (B) 8 (C) 12 (D) 16

Question 14 (JAM 2007 • Q6 • ID: JAM_2007_Q6) MCQ +6 M Neg: −2.0

6. Let G be an Abelian group of order 10. Let $S = \{g \in G : g^{-1} = g\}$. Then the number of non-identity elements in S is

(A) 5
(B) 2
(C) 1
(D) 0

Question 15 (JAM 2006 • Q13 • ID: JAM_2006_Q13) MCQ +6 M Neg: −2.0

13. Let G be a group of order 7 and $\phi(x) = x^4$, $x \in G$. Then ϕ is

(A) not one – one
(B) not onto
(C) not a homomorphism
(D) one – one, onto and a homomorphism

Question 16 (JAM 2005 • Q8 • ID: JAM_2005_Q8) **MCQ** **+6 M** **Neg: −2.0**

8. In the group $\{1, 2, \ldots, 16\}$ under the operation of multiplication modulo 17, the order of the element 3 is

(A) 4

(B) 8

(C) 12

(D) 16

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Groups (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	**D**
2	MCQ	+2	**A**
3	MCQ	+2	**B**
4	MCQ	+1	**B**
5	MCQ	+2	**A**
6	MCQ	+6	**B**
7	MCQ	+6	**C**
8	MCQ	+6	**B**

Q. No.	Type	Marks	Official Key
9	MCQ	+6	**A**
10	MCQ	+6	**D**
11	MCQ	+6	**C**
12	MCQ	+6	**C**
13	MCQ	+6	**B**
14	MCQ	+6	**C**
15	MCQ	+6	**D**
16	MCQ	+6	**D**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.